How Bike Derailleurs Work

01 Where does the chain travel?



The chain loops around a front chainring and one cog in the back cassette.

Look closer. Follow the chain around the front chainring, back to one cog in the cassette, and through the two small pulley wheels below.

A little explanation. The chain forms one long loop. It grips teeth on a front gear, called a chainring, and on one back gear, called a cog. Turning the pedals pulls the chain and turns the back wheel.

Try this. Use a finger in the air to trace the whole loop without touching the bike. Name the front chainring, back cog, and lower pulley wheels.

02 What does the rear derailleur move?



Cable pull and spring return move the rear derailleur sideways from cog to cog.

Look closer. Look below the back cogs for the small arm that lines its upper pulley with the cog holding the chain.

A little explanation. On a mechanical bike, the shifter pulls or lets go of a cable. The cable and a return spring move the rear derailleur sideways. Its upper pulley guides the moving chain from one back cog to the next.

Try this. Hold two hands apart like neighboring cogs. Move one finger sideways from one hand to the other to model the chain's short trip.

03 Why are there two small pulley wheels?



One pulley guides the chain while the other moves to take up slack.

Look closer. See the chain bend around an upper guide pulley and a lower tension pulley before it heads forward again.

A little explanation. Different cogs need slightly different amounts of chain. A spring turns the derailleur cage, and the lower pulley moves with it to take up slack. The upper pulley guides the chain toward the chosen cog.

Try this. Curve a piece of yarn around two bottle caps on a table. Move the lower cap gently and notice how the extra loop becomes longer or shorter.

04 Why do big and small back cogs feel different?



A big back cog makes hills easier; a small back cog gives more wheel turns on flat ground.

Look closer. Compare the largest back cog with the smallest, then find the teeth that catch the chain.

A little explanation. A big back cog is an easy gear for hills: the wheel turns less for each pedal turn, so the pedals need less push. A small back cog is a hard gear that can carry the bike farther per pedal turn on flat ground. With 32 teeth in front and 16 in back, the wheel turns 2 times for one pedal turn.

Try this. With an adult, place the bike upside down on a flat floor or secure it in a stand. Turn one pedal slowly forward by hand while only the adult shifts. Watch the derailleur move the chain. Count teeth on the smallest and biggest back cogs, then count wheel turns for one pedal turn in an easy gear and a hard gear. Keep fingers away from the chain, gears, and spokes whenever anything is turning. No riding is needed.